The empirical recurrence for OEIS A229517, recovered from its tail

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Abstract

OEIS A229517 records "Empirical recurrence of order 75" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 122$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 78$. Its last coefficient is $c_{75} = 59 \neq 0$, so no further tail satisfies anything shorter; any order-75 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A229517 is "Number of n X 4 0..2 arrays of the median of the corresponding element, the element to the east and the element to the south in a larger (n+1) X 5 0..2 array without adjacent equal elements in the latter.". It has offset 1 and begins

$$39, 481, 4193, 34151, 275243, 2223913, 17910863, 144971845, \ \ldots$$

The entry states:

> Empirical recurrence of order 75 (see link above).

As of the "Last modified" line on the live entry (Apr 27 2021 21:05:51, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 122$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 122$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 75: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 75.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 244$ exact terms from $n = 3$ on, it returns order 75 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{75} c_i \, a(n-i),$$

c_1	2	c_{20}	-9153391	c_{39}	-356179595	c_{58}	19958745
c_2	69	c_{21}	106009883	c_{40}	313637920	c_{59}	-10388587
c_3	-47	c_{22}	-12903083	c_{41}	-216778087	c_{60}	-27867337
c_4	-1110	c_{23}	42643069	c_{42}	232914149	c_{61}	5130499
c_5	1048	c_{24}	122555060	c_{43}	275953328	c_{62}	4625020
c_6	5026	c_{25}	-265259912	c_{44}	-87106945	c_{63}	-18379
c_7	-7686	c_{26}	-272726253	c_{45}	-10197531	c_{64}	1191516
c_8	12236	c_{27}	-32825170	c_{46}	-151234981	c_{65}	444434
c_9	17136	c_{28}	193763131	c_{47}	-66598994	c_{66}	-891330
c_{10}	-74253	c_{29}	-203185205	c_{48}	-29638060	c_{67}	-130540
c_{11}	105726	c_{30}	240816576	c_{49}	118460011	c_{68}	103682
c_{12}	-258563	c_{31}	-172323396	c_{50}	171470888	c_{69}	-82438
c_{13}	-1432462	c_{32}	-304487274	c_{51}	-138709641	c_{70}	-43509
c_{14}	2665405	c_{33}	-21071932	c_{52}	15648487	c_{71}	24091
c_{15}	-479114	c_{34}	14675013	c_{53}	-35040625	c_{72}	5225
c_{16}	-1565367	c_{35}	70690284	c_{54}	-106407781	c_{73}	-1393
c_{17}	-2741221	c_{36}	300578604	c_{55}	54785836	c_{74}	-357
c_{18}	1171931	c_{37}	312311811	c_{56}	18564181	c_{75}	59
c_{19}	3919563	c_{38}	-107305713	c_{57}	-7384687		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 78$, and it is the recurrence OEIS A229517 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{75} - c_1 x^{75-1} - \cdots - c_{75}$. Its constant term is $-c_{75} = -59$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 75 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 75, so $\deg q = 75$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 75, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 75, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 59.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-75 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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